POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks:100
Course: Calculus I | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) State Leibnitz theorem. If y=$\left(x+\sqrt{1+x^2}\right)^m$, show that 8

i. $\left(1+x^2\right)y_2+xy_1-m^2y=0$

ii. $\left(1+x^2\right)y_{n+2}+(2n+1)xy_{n+1}+\left(n^2-m^2\right)y_n=0$

b) State Rolle's Theorem and write its geometrical meaning. Verify it for $f(x)=(x-a)^m(x-b)^n,\ x\epsilon[a,b]$ where m and n are positive integers. 7

OR

b) Assuming the validity of the expansion, expand $f(x)=\tan x$ using Maclaurin's series and hence the expansion of $\sec^2 x$.

2. a) Find the asymptotes of the curve $(x-y)^2x^2-a^2\left(x^2+y^2\right)=0$. 8

b) Find the radius of curvature at any point of the parametric curve $x=a(t+\sin t), y=a(1-\cos t)$. 7

3. Integrate **Any Three** of the following: 3×5

a) $\int\frac{1}{2-3\sin 2x}dx$

b) Prove that: $\int_0^{\pi/2}\frac{\sqrt{\cot x}}{1+\sqrt{\cot x}}dx=\frac{\pi}{4}$

c) Evaluate: $\int_1^{\infty}\frac{x}{(1+x^2)^2}dx$

d) By using gamma function evaluate $\int_0^{\frac{\pi}{6}}\sin^4 3x\cos^2 6x dx$

4. a) Find the volume of the solid in the region in the first quadrant bounded by the curve $y=x^2$, and the lines $y=0, x=1$ about the line $x=-1$. 8

b) State and prove Euler's theorem of homogenous function of two variables x and y of degree n. 7

If $u = sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}tanu$

5. a) Use Lagrange's Multipliers Method, Find the minimum value of $f = x^2 + y^2 + z^2$ such that $x + y + z = 3a^2$ 8

b) Define Bernoulli's equation. Solve: $\frac{dy}{dx} - y\tan x = -y^2 \sec x$. 7

OR

b) A tank initially contains 40 kg of salt dissolved into 200 liter of water. A solution of 2 kg of salt per liter is allowed to enter the tank at the rate of 5 liters per minute and uniform solution is drained from the tank at the same rate. Find the amount of salt at any time t. Also determine the salt in tank in 15 minutes.

6. a) Define Euler -Cauchy equation. Solve the initial value problem: 8

$$4x^2y'' + 24xy' + 25y = 0, y(1) = 2, y'(1) = -6$$

b) Solve by method of variation of parameter $y'' - 2y' + y = \frac{e^x}{x^3}$ 7

7. Attempt all Questions. 4×2.5

a) If function $f(x) = \begin{cases} 2 - x & for\ 1 < x < 2 \\ x - \frac{x^2}{2} & for\ x \geq 2 \end{cases}$

Examine continuity of a function at x=2.

b) Check line of symmetry of the curve $y^2(a-x) = x^2(a-x)$.

c) Find the arc length of the curve $x^2+y^2=a^2$

d) Solve: $(1 + x)y\,dx + (1 + y)x\,dy = 0$.